

Synthetic Corrosion Inspection Report

SYNTHETIC DEMO DATA - NOT AN OPERATING RECORD

Facility	Kavach Demonstration Refinery
Unit	Crude Distillation Unit (CDU)
Equipment	Tower T-101
Inspection date	10 September 2026
Inspection method	Visual inspection and ultrasonic thickness checks

Inspection summary

This demonstration report simulates a routine integrity review. It is designed for Kavach AI document extraction, knowledge-grounding, and approval-note workflows.

Key findings

- Shell thickness at the lower course measured 12.5 mm; minimum allowable thickness is 10.0 mm.
- Minor localized pitting was observed near nozzle N-1. No active leak was observed.
- Overhead condenser piping showed early chloride-attack indicators requiring process review.

Recommended actions

- Schedule focused ultrasonic testing for nozzle N-1 within 90 days.
- Verify wash-water injection rate and review sour-water chloride trend with Operations.
- Record photographs and thickness readings in the next integrity review pack.

Demo task

Extract the findings, compare them against the local corrosion-monitoring SOP, and generate a concise approval note for the inspection manager.